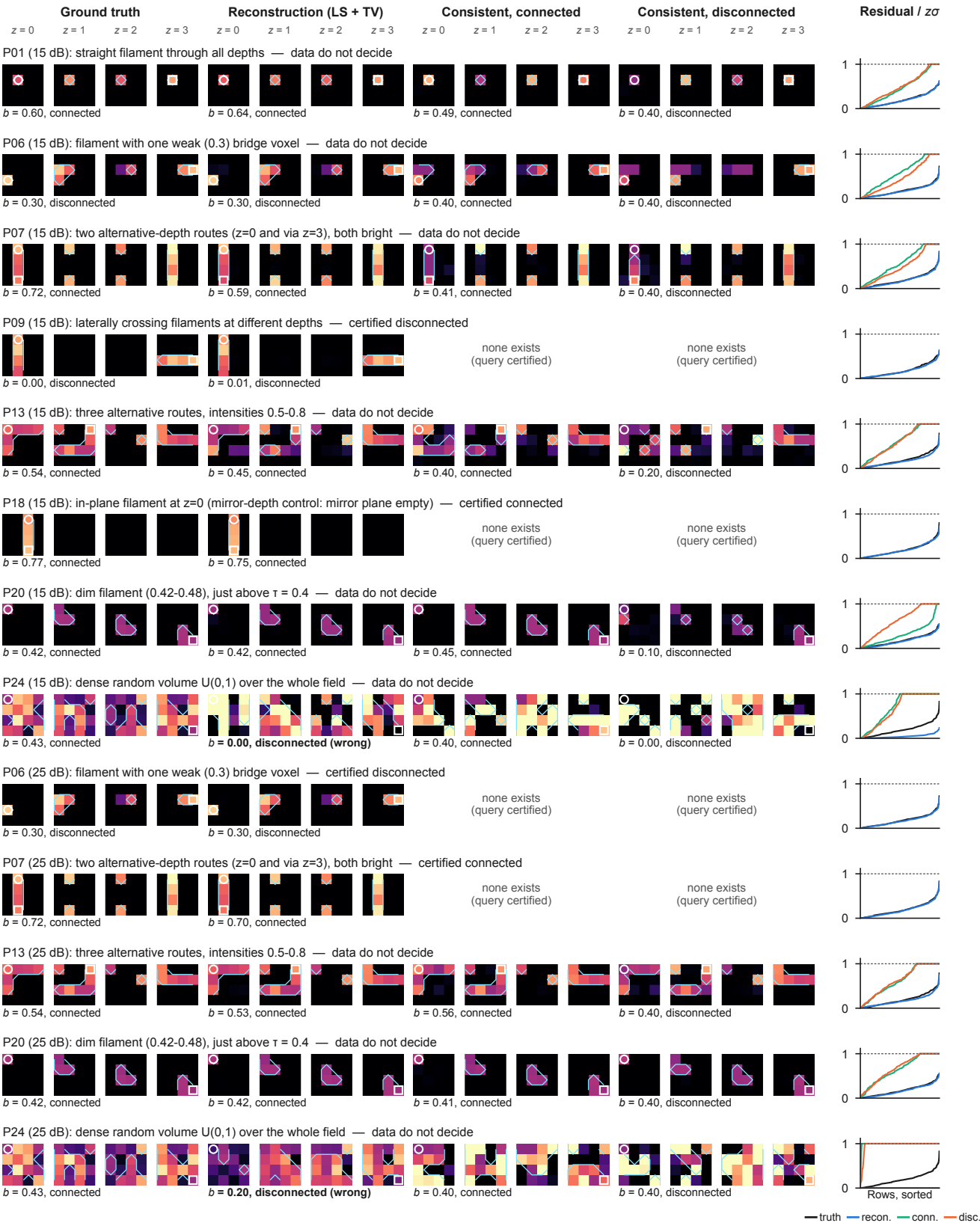

Ground truth
Reconstruction (LS + TV)
Consistent, connected
Consistent, disconnected
Residual / zσ
z = 0
z = 1
z = 2
z = 3
P01 (15 dB): straight filament through all depths — data do not decide
b = 0.60, connected
b = 0.64, connected
b = 0.49, connected
b = 0.40, disconnected
P06 (15 dB): filament with one weak (0.3) bridge voxel — data do not decide
b = 0.30, disconnected
b = 0.30, disconnected
b = 0.40, connected
b = 0.40, disconnected
P07 (15 dB): two alternative-depth routes (z=0 and via z=3), both bright — data do not decide
b = 0.72, connected
b = 0.59, connected
b = 0.41, connected
b = 0.40, disconnected
P09 (15 dB): laterally crossing filaments at different depths — certified disconnected
b = 0.00, disconnected
b = 0.01, disconnected
none exists (query certified)
none exists (query certified)
P13 (15 dB): three alternative routes, intensities 0.5-0.8 — data do not decide
b = 0.54, connected
b = 0.45, connected
b = 0.40, connected
b = 0.20, disconnected
P18 (15 dB): in-plane filament at z=0 (mirror-depth control: mirror plane empty) — certified connected
b = 0.77, connected
b = 0.75, connected
none exists (query certified)
none exists (query certified)
P20 (15 dB): dim filament (0.42-0.48), just above τ = 0.4 — data do not decide
b = 0.42, connected
b = 0.42, connected
b = 0.45, connected
b = 0.10, disconnected
P24 (15 dB): dense random volume U(0,1) over the whole field — data do not decide
b = 0.43, connected
b = 0.00, disconnected (wrong)
b = 0.40, connected
b = 0.00, disconnected
P06 (25 dB): filament with one weak (0.3) bridge voxel — certified disconnected
b = 0.30, disconnected
b = 0.30, disconnected
none exists (query certified)
none exists (query certified)
P07 (25 dB): two alternative-depth routes (z=0 and via z=3), both bright — certified connected
b = 0.72, connected
b = 0.70, connected
none exists (query certified)
none exists (query certified)
P13 (25 dB): three alternative routes, intensities 0.5-0.8 — data do not decide
b = 0.54, connected
b = 0.53, connected
b = 0.56, connected
b = 0.40, disconnected
P20 (25 dB): dim filament (0.42-0.48), just above τ = 0.4 — data do not decide
b = 0.42, connected
b = 0.42, connected
b = 0.41, connected
b = 0.40, disconnected
P24 (25 dB): dense random volume U(0,1) over the whole field — data do not decide
b = 0.43, connected
b = 0.20, disconnected (wrong)
b = 0.40, connected
b = 0.40, disconnected
1
0
Rows, sorted
truth
recon.
conn.
disc.